Probabilistic (Graphical) Models

Part 2: Probabilities and Bayesian Learning/Inference

2025

Rémi Emonet

PGM M2 MLDM

Université Jean Monnet – Laboratoire Hubert Curien

'a'/'z' keys to browse the slides, 'o' for an overview, 'd' for dark/light mode

Objectives for the (P)(G)(M) Course

- Understanding
  - what is a probabilistic model
    - from fitting a distribution
    - to huge generative models
  - what is a Probabilistic Graphical Model (PGM)
  - what are the inference/learning methods
  - how this relates/differs/interacts with other approaches (e.g. deep learning, VAE, diffusion)
- Skills, be able to
  - model a problem with probabilities
  - reason using probabilities
  - read and write graphical models
  - select (and implement) models and inference methods

Syllabus for the PGM Course / Disclaimer

α. Introduction, the need for uncertainty

β. Probabilistic/generative modelling, maximum likelihood

γ. Probabilities, Bayesian principles and Bayesian learning

δ. Closed-form learning (conjugate priors)

ε. Inference Methods for Probabilistic (Latent Variable) Models

ζ. Variational Auto-Encoders (and variational inference)

η. Deep Learning meets Probabilistic Models

θ. Student Project

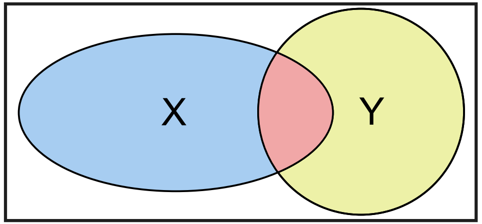
All you need to know about know about probabilities (almost)

What is a probability?

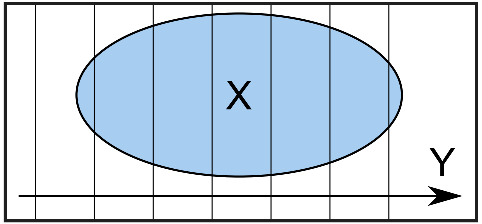
✐

Probabilities and Measure Theory

- Product rule
  - $p(X,Y) = p(X|Y)\,p(Y) = p(Y|X)\,p(X)$

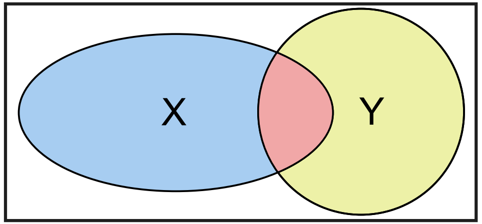


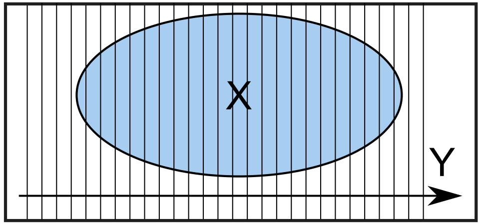
- Marginalization, "Sum rule"
  - $p(X) = \sum_Y p(X,Y) \triangleq \sum_y p(X, Y=y)$



- Bayes rule
  -
  - $p(Y|X) = \frac{p(X|Y)\,p(Y)}{p(X)}$
  - $p(X) = \sum_Y p(X|Y)\,p(Y) \triangleq \sum_y p(X|Y=y)\,p(Y=y)$
- NB: Expectation
  - $\mathbb{E}[Y] = \sum_y p(Y=y)y$
  - $F = f(Y)$ a random variable
  - $\mathbb{E}[F] = \mathbb{E}_{y\sim p(Y)}[f(Y)] \triangleq \sum_y p(Y=y)f(y)$ LOTUS

Discrete and Continuous Random Variables





- Some discrete distributions
  - bernoulli, $p(X == 0) = 1 - \theta$ and $p(X == 1) = \theta$
  - binomial
  - categorical
  - multinomial
  - poisson
- Some continuous distributions
  - normal distribution $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
  - beta, laplace, gamma, …
- Same “rules” with continuous probability density functions
  - product rule: $f_{X,Y}(x, y) = f_{X|Y}(x, y) f_Y(y)$
  - "sum rule" (marginalization): $f_X(x) = \int f_{X,Y}(x, y) dy = \int f_{X|Y}(x|y) f_Y(y) dy$
  - Bayes’ rule: $f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y) f_Y(y)}{f_X(x)}$
  - NB: we’ll still use "p()" + see also probability review
  - LOTUS : $\mathbb{E}[F] = \mathbb{E}_{y \sim p(Y)}[f(Y)] \triangleq \int p(Y = y) f(y) dy \cdots\cdots\cdots \triangleq \int_\Omega f \circ Y \, d\mu \triangleq \int_{\Omega'} g \, d(Y_{\#\mu})$

Bayesian Principles and Learning

What is it to be Bayesian?

- Do you wanna flip a coin?
- Setting (we know flips are independent)
  - 3H 5T
  - 1M€ on TTTTT
  - How much should you bet?
    - 32 258 €? with $\theta = 50/50$
    - 105 461 €? with $\theta = 5/8$ (following MLE)
- Issue with MLE?
- Issue with $\theta = \ldots$?

Learning distribution parameters

- What should we estimate or optimize?
  - $p(X|\theta)$?
  - $p(\theta|X)$?
  - $p(X,\theta)$?
  - what the Bayes' rule tells us? $$p(\theta|X) = \frac{p(X|\theta)p(\theta)}{p(X)}$$
    - what is $p(\theta)$?
    - what is $p(X)$?
- MLE
  - $\text{argmax } p(X|\theta)$
  - $\text{argmax } \log p(X|\theta)$
- MAP (maximum a posteriori)
  - $\text{argmax } \log p(\theta|X)$
  - $\text{argmax } \log(p(X|\theta)p(\theta)/p(X))$
  - $\text{argmax } \log p(X|\theta) + \log p(\theta) - \cancel{\log p(X)}$

Bayesianism

Bayesian (people)

1. use probabilities to represent (lack of) knowledge
2. have (explicit) priors on everything unknown
3. keep distributional estimate as long as possible/needed

Introduction to learning/inference

Common Tasks/Problems around Probabilistic Models

- Identify the (random) variables of interest
- Model the joint distribution of these variables $p(X)$ (or $p(X, Y)$)
  - what structure has $p(X)$
  - what knowledge do we want to put in
- Learn/Estimate the density $p(X)$ from data
  - anomaly detection: x_{ano} is an anomaly $p(X = x_{ano})$ is low
  - model selection: $p_{mod1}(X) > p_{mod2}(X)$?
- Do inference/reasoning, from a given/learned model
  - with observed/fixed variables, $p(X_j | X_i = 42)$
  - with missing variables, $p(X_j) = \sum_v p(X_j, X_i = v)$
- Generate new samples from $p(X)$, i.e., $x_{new} \sim p(X)$

Examples

- Text
  - language modeling
  - translation
  - text generation
- Image / video
  - generation ☺ ⛰️ 🐈 🪑
  - inpainting
  - denoising
  - super-resolution
  - video tracking
- Audio
  - speech recognition (speech to text)
  - speech synthesis (text to speech)
- And all sciences: biology, medicine, ecology, social sciences, …

Inference vs Learning

- Inference
  - finding the most probable values for latent/hidden variables
  - (given some parameters and observations)
- Learning
  - finding the “best” parameters
  - given some observation
- In a more “Bayesian” world
  - we put priors on everything
  - the parameters of interest become latent variables
  - learning = inference
- In a “full Bayesian” world
  - keep the full distribution (a distributional estimate)
  - not just the best/most probable

Coin Flipping: Code-assisted Bayesian Treatment

- Do you wanna flip a coin?
- ✏️ ⌨️
  - discretize everything
  - multiply prior and likelihood
  - normalize to get posterior

Closed-Form Bayesian Learning

Do you wanna flip a coin? again? From MLE to MAP

$$
\begin{aligned}
\log p(X|\theta) + \log p(\theta) &= \log \prod_{i=1}^{N} p(x_i|\theta) + \log p(\theta) && (1)\\
&= \sum_{i=1}^{N} \log p(x_i|\theta) + \cdots && (2)\\
&= \sum_{i=1}^{N} \log \theta^{x_i}(1-\theta)^{1-x_i} + \cdots && (3)\\
&= \sum_{i=1}^{N} x_i \log \theta + (1-x_i)\log(1-\theta) + \cdots && (4)\\
&= N_h \log \theta + N_t \log(1-\theta) + \cdots && (5)\\
&= \cdots && (6)
\end{aligned}
$$

Conjugate Priors

- We want a prior $p(\theta)$ that makes it easy to compute the posterior $p(\theta|X)$

- Conjugate prior
  - a prior $p(\theta)$ is conjugate to the likelihood $p(X|\theta)$
  - if the posterior $p(\theta|X)$ is in the same family as the prior

- Coin flipping example
  - likelihood $p(X|\theta)$ is a bernoulli/binomial distribution
  - conjugate prior $p(\theta)$ is a beta distribution
    - $p(\theta) = \mathrm{Beta}(a, b)(\theta) = \frac{1}{B(a,b)} \theta^{a-1} (1-\theta)^{b-1}$
    - $B(a, b)$ is a normalizing constant $B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$
    - a and b are prior hyper-parameters, interpretation?
  - posterior $p(\theta|X)$ is also a beta distribution

Posterior Predictive and Conjugate Priors

- Posterior predictive
  - predict a new observation x_{new} given past observations X
  - the probability of a new observation x_{new} under the posterior distribution $p(\theta|X)$

$$p(x_{new}|X) = \int p(x_{new}|\theta, X) p(\theta|X) d\theta$$

$$p(x_{new}|X) = \int p(x_{new}|\theta) p(\theta|X) d\theta \quad \text{(indep.)}$$

- Coin flipping example (with conjugacy)

$$p(x_{new} = 1|X) = \int_0^1 p(x_{new} = 1|\theta) p(\theta|X) d\theta = \int_0^1 \theta \cdot \mathrm{Beta}(N_h + a, N_t + b)(\theta) d\theta$$

$$p(x_{new} = 1|X) = \mathbb{E}\left[Beta(N_h + a, N_t + b)\right] = \frac{N_h + a}{N_h + N_t + a + b}$$

- More in the Wikipedia huge table

Overview

- All you need to know about know about probabilities (almost)
- Bayesian Principles and Learning
- Introduction to learning/inference
- Closed-Form Bayesian Learning

Next

- Learning without closed form

For next time:

- Be clear on all these Bayesian stuff: prior, posterior, likelihood, conjugacy, posterior predictive
- Find the conjugate prior for a gaussian with known variance (and unknown mean)
- Look at what is a mixture distribution / model (https://en.wikipedia.org/wiki/Mixture_distribution)
- Challenge:
  > - Start trying to find a closed-form MLE for a mixture of 2 gaussians (with known variance)!
  > - How could you simplify the problem so it can be solved in closed form?